The first joint Hamiltonian layers above a Slater state

Same-disorder mixed fourth moments in complex double-scaled SYK

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Abstract

The Slater-adapted creation block of the original Hamiltonian has a dressed bosonic vacuum law, but captures only part of the first Hamiltonian excitation. We retain its correlations with the same Hamiltonian and evaluate every joint word through degree four. Finite-size Gaussian pairings reduce to elementary counts and the known embedded-ensemble covariance channel. Their double-scaled limit gives a positive seven-dimensional space of polynomial vectors of degree at most two. The first Hamiltonian actions exhibit a new commutator direction of squared norm $1-q$. The actual disorder complement has normalized fourth moment $2+\sqrt{q}+2q$; replacing it by another standard q-Gaussian mode would change the model. These results determine the first layers, with the all-degree joint representation and its bulk interpretation left as the next tasks.

1 State, normalization, and the literature interface

Use standard complex CAR, $\{c_i, c_j^\dagger\} = \delta_{ij}$, and

$$H_N = \sqrt{w_N} \sum_{|A|=|B|=p} J_{AB} C_A^\dagger C_B, \quad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p}\right)^2, \quad \lambda_N = \frac{p^2}{N}. \tag{1}$$

Here $C_A^\dagger$ has increasing site order, C_A is its adjoint, and the centered Hermitian Gaussian coefficients obey $\mathbb{E} J_{AB} J_{CD} = \delta_{AD}\delta_{BC}$. Overlapping A, B are included. Fix even N, $m = a = N/2$, an occupied set O of size m, its empty complement E, and $\Omega_O = |O\rangle$. All moments use

$$\varphi_N(X) = \mathbb{E}\langle \Omega_O, X\Omega_O\rangle. \tag{2}$$

This is the annealed Slater vector state. It is not assumed stationary. Take $p \to \infty$, $\lambda_N \to \lambda > 0$, fixed $J_s > 0$ and fixed word length, with $\hbar = 1$ and string-energy units for H_N. The normalized h_N below is dimensionless.

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The same-J excitation block and normalizations are

$$R_N^\dagger = \sum_{A\subset E,\ B\subset O} J_{AB} C_A^\dagger C_B, \qquad L_N = \binom{m}{p}\binom{a}{p}, \tag{3}$$

$$D_N = \binom{m}{p}\binom{a+p}{p}, \qquad h_N = \frac{H_N}{\sqrt{w_N D_N}}, \qquad M_N^\dagger = \frac{R_N^\dagger}{\sqrt{L_N}}. \tag{4}$$

Every subset in the sum has size p; M_N is the adjoint. Set

$$q = e^{-4\lambda}, \quad s = \sqrt{q}, \quad \gamma = q^{1/4}, \quad \gamma_N = \sqrt{L_N/D_N}, \quad \delta = \sqrt{1-s}. \tag{5}$$

The preceding stage proved the pure-M fixed-word law $M = q^{n/2}b$, $M^\dagger = b^\dagger q^{n/2}$ in its vacuum state, and the first-step fraction $L_N/D_N \to s$. Here we retain the rest of $h_N\Omega_O$.

Rajgadia–Xu's state-adapted construction uses matter Gaussian coefficients separate from those of H; see their HTML Eqs. (42)–(44) and the selection rule after (47) [1]. That selection rule is not an input for the present same-J model. Already at degree two,

$$\varphi_N(h_N^2) = \varphi_N(M_N M_N^\dagger) = 1, \qquad \varphi_N(h_N M_N^\dagger) = \varphi_N(M_N h_N) = \gamma_N. \tag{6}$$

All other two-letter moments vanish. Odd total degree vanishes by Gaussian parity, but unequal numbers of M and $M^\dagger$ need not make a mixed word vanish.

2 An exact finite-size table for the three pairings

In this section only, use the raw alphabet $H, +, -$ for $H_N/\sqrt{w_N}, R_N^\dagger, R_N$. Omit subscripts on counts. A Gaussian edge with two H endpoints is type G; every other edge is type S, restricted to the excitation block or its adjoint. An S edge is forward if its creation acts first (right to left). An endpoint $+$ is a creation and $-$ is an annihilation, fixing the opposite endpoint even when that endpoint is H. Two equal non-H endpoints are incompatible and give zero.

For four written slots $0, 1, 2, 3$, the three pairings are

$$(01)(23) \text{ disjoint}, \qquad (02)(13) \text{ crossed}, \qquad (03)(12) \text{ nested}. \tag{7}$$

In a nested pairing the first edge listed is outer. Define

$$r_N = \frac{\binom{m-p}{p}\binom{a-p}{p}}{L}, \qquad T_N = \binom{m-p}{p}\binom{a}{p}, \tag{8}$$

$$W_k = \binom{m}{p}\binom{p}{k}\binom{a}{k}, \qquad U_N = \sum_{k=0}^{p} W_k \binom{a-k}{p}\binom{m-k}{p}. \tag{9}$$

Binomials outside their natural range are zero. K_N denotes the raw crossed $HHHH$ contraction.

Edges / orientation	Disjoint	Crossed	Nested
G, G	D^2	K_N	D^2
S, S, both forward	L^2	$L^2 r_N$	$L^2 r_N$
G, S, S forward	DL	LT_N	see below
S outer, G inner	—	—	DL
G outer, S inner forward	—	—	U_N
S outer forward, S inner backward	0	0	L
G outer, S inner backward	0	0	L

All other backward cases are zero. Sum the three contractions, then divide by $D^{n_H/2} L^{(4-n_H)/2}$ to obtain the normalized moment. This specifies all 3^4 ordered fourth words.

Derivation of the entries

For a fixed excitation label $\alpha = (A, B)$, write $X_\alpha = C_A^\dagger C_B$. Relative to Ω_O, this is a pure creation monomial of even degree $2p$. Two active excitation labels must be disjoint on both O and E. Their even CAR monomials then commute, and each creation/adjoint pair has amplitude $+1$. Disjoint time intervals give L^2; overlapping intervals give $L\binom{m-p}{p}\binom{a-p}{p}$. If an inner selected edge is backward, it can annihilate an outer selected excitation only by taking the same entire label. It then recreates it, giving exactly L. All other backward placements meet the vacuum boundary or cannot remove the required particles and holes.

For general labels $Y_{AB} = C_A^\dagger C_B$, the channel

$$\Phi(X) = \sum_{A,B} Y_{AB} X Y_{AB}^\dagger \tag{10}$$

has $\Phi(I) = DI$ on the m-particle sector. Thus adjacent G contractions and G nested inside a selected edge give the table's D factors. The established embedded-ensemble spectrum is

$$\Lambda^r(p) = \binom{m-r}{p}\binom{N-m+p-r}{p} \tag{11}$$

on rank-r irreducible tensors [2, Eqs. (17),(25)–(26)]. A disjoint p-hop X_α is a rank-p tensor: choose its creation and annihilation indices as opposite extremal sets to obtain a highest-weight vector; site permutations generate every such disjoint hop. Hence $\Phi(X_\alpha) = T_N X_\alpha$, giving the crossed GS value LT_N. This is an application of the known channel, not a newly discovered covariance eigenvalue.

For G outer and S inner, let k be the number of particles that the first general hop moves from O to E. The exact number of allowed raw labels is W_k. Indeed choose its p annihilated occupied sites, the k particles left displaced among those p, and their k empty destinations; the other $p-k$ are restored. After this hop there are $a-k$ empty E sites and $m-k$ filled O sites. A forward selected pair therefore has $\binom{a-k}{p}\binom{m-k}{p}$ choices. Summing gives U_N. A backward selected pair requires $k=p$, and must undo the whole first hop, giving $W_p = L$. These are sums of squared CAR amplitudes; their signs are not discarded by an absolute-value approximation.

For completeness the standard finite G, G expression is

$$K_N = \frac{1}{\binom{N}{m}} \sum_{r=0}^{\min(p,m-p)} \Lambda^r(m-p)\Lambda^r(p) \left[\binom{N}{r}^2 - \binom{N}{r-1}^2 \right], \tag{12}$$

with $\Lambda^r(k) = \binom{m-r}{k}\binom{N-m+k-r}{k}$; this is the crossed term in the existing fourth-moment formula [2, Eq. (28)]. Alternatively K_N can be left as the raw CAR sum. Its double-scaled ratio $K_N/D_N^2 \to q$ is the ordinary complex DSSYK fourth-moment input already used in the preceding stages [3, Sec. 5.1]. Unitary invariance makes the annealed H-only Slater moments equal to the sector trace; the half-filled sector has intersection parameter $e^{-4\lambda}$. No finite identity inferred from a few fixtures is needed here.

3 Double-scaled mixed fourth moments

Elementary binomial ratios and the preceding first-step deficit law give

$$\frac{L}{D} \to s, \qquad r_N \to q, \qquad \frac{T_N}{D} \to q, \qquad \frac{U_N}{DL} \to q. \tag{13}$$

For example, for fixed c,

$$\log \frac{\binom{m-cp}{p}}{\binom{m}{p}} = \sum_{j=0}^{p-1} \log\left(1 - \frac{cp}{m-j}\right) = -\frac{cp^2}{m} + o(1). \tag{14}$$

For the last ratio in (13), view W_k/D as a probability. The deficit $p-k$ converges to Poisson(2λ) from the preceding stage. On any fixed deficit window the remaining binomial product divided by L converges uniformly to q. That product lies in $[0,1]$, so tightness controls the window's complement. This proves the limit without interchanging an uncontrolled tail and the large-N limit. The nonzero backward terms vanish after normalization (as $1/L$ or $1/D$, with bounded mixed factors).

One can now evaluate each limiting diagram with endpoint factors 1 for HH or $MM^\dagger$, and γ for $HM^\dagger$ or MH. Reject backward or incompatible selected edges. Every crossing contributes q. A nesting contributes q precisely when the inner edge is selected. All disjoint diagrams and the remaining nesting have factor one. This is a *two-edge rule*; an arbitrary-edge version has not been proved. Some useful moments are

$$\varphi(h^4) = 2 + q, \qquad \varphi(h^3 M^\dagger) = \gamma(2+q), \tag{15}$$

$$\varphi(h^2 (M^\dagger)^2) = 2sq, \qquad \varphi(hM^\dagger h M^\dagger) = s. \tag{16}$$

Here and below limiting notation abbreviates limits of the specified finite sequences, not an assumed full representation.

4 A positive space and the first Hamiltonian actions

For polynomials P, Q of degree at most two, set $\langle P\Omega, Q\Omega\rangle = \lim_N \varphi_N(P_N^\dagger Q_N)$. Every entry exists by the table, and positivity follows from finite-N positivity. Quotient by null vectors. The identities $M\Omega = 0$, $MM^\dagger\Omega = \Omega$ in this quotient follow from the preceding pure-ladder law. The additional relation is

$$Mh\Omega = \gamma\Omega, \qquad \mathbb{E}\|(M_N h_N - \gamma_N I)\Omega_O\|^2 = \frac{1}{D_N}. \tag{17}$$

Thus words ending in M vanish on the vacuum, and the remaining degree-two space is spanned by $I, M^\dagger, h, (M^\dagger)^2, M^\dagger h, hM^\dagger, h^2$ acting on Ω.

Define the following four centered degree-two vectors:

$$U = (M^\dagger)^2\Omega, \tag{18}$$

$$V = \left(M^\dagger h - \gamma (M^\dagger)^2\right)\Omega, \tag{19}$$

$$D_v = \left(hM^\dagger - M^\dagger h - \gamma I\right)\Omega, \tag{20}$$

$$R_v = (h^2 - I)\Omega - sU - 2\gamma V - \gamma D_v. \tag{21}$$

They are mutually orthogonal and orthogonal to Ω and all degree-one vectors. Their norms are

$$\begin{aligned} &\|U\|^2 = 2q, \qquad \|V\|^2 = q(1-s), \qquad \|D_v\|^2 = 1 - q, \\ &\|R_v\|^2 = (1-s)^2(1+s+2q). \end{aligned} \tag{22}$$

For example $\|hM^\dagger\Omega\|^2 = 1 + s(1+q)$, while $\|M^\dagger h\Omega\|^2 = \langle hM^\dagger\Omega, M^\dagger h\Omega\rangle = q(1+s)$; these give the D_v norm after subtracting its vacuum component. All other entries follow from the same table. Gaussian parity sets inner products with degree-one vectors to zero.

Let $e_1 = M^\dagger\Omega$, $e_r = (h\Omega - \gamma e_1)/\delta$, and normalize the four vectors as

$$e_2 = \frac{U}{\sqrt{2q}}, \quad e_{r2} = \frac{V}{\delta\sqrt{q}}, \quad e_d = \frac{D_v}{\sqrt{1-q}}, \quad e_{rr} = \frac{R_v}{(1-s)\sqrt{1+s+2q}}. \tag{23}$$

For fixed $0 < q < 1$ every denominator is strictly positive. The seven vectors $\Omega, e_1, e_r, e_2, e_{r2}, e_d, e_{rr}$ are an orthonormal basis of this degree-at-most-two quotient. Multiplication by h is determined from degree at most one into this space:

$$h\Omega = \gamma e_1 + \delta e_r, \tag{24}$$

$$he_1 = \gamma\Omega + \gamma\sqrt{2q}\, e_2 + \delta\sqrt{q}\, e_{r2} + \sqrt{1-q}\, e_d, \tag{25}$$

$$he_r = \delta\Omega + \gamma\sqrt{q}\, e_{r2} + \delta\sqrt{1+s+2q}\, e_{rr}. \tag{26}$$

These follow by substituting the four definitions; (17) ensures the quotient relation needed below. No action of h on the degree-two basis is specified. Consequently this is not a closed seven-state Hamiltonian or a finite-time solution.

The extra commutator direction has a particularly simple test:

$$\|([h, M^\dagger] - \gamma I)\Omega\|^2 = 1 - q > 0. \tag{27}$$

It is orthogonal to the pure second ladder state e_2, to the mixed direction e_{r2}, and to the residual pair direction e_{rr}. This resolves a definite part of the Hamiltonian leakage instead of treating all missing norm as one scalar correction.

5 The actual Gaussian-disorder complement

At finite N define

$$g_N = h_N - \gamma_N(M_N + M_N^\dagger). \tag{28}$$

This removes exactly the selected excitation block and its adjoint from the original Hamiltonian. The remaining Gaussian coordinates are independent of the selected coordinates. That statement is about disorder coordinates; the CAR matrix factors still share the same quantum state. The table gives

$$\varphi(g^2) = 1 - s, \quad (g^2 - (1-s)I)\Omega = R_v, \quad \frac{\varphi(g^4)}{\varphi(g^2)^2} = 2 + s + 2q. \tag{29}$$

To see the vector identity, expand $g^2\Omega$ using $M\Omega = 0$, $MM^\dagger\Omega = \Omega$, and (17); the result is exactly the definition of R_v plus $(1-s)\Omega$. Its norm then yields (29). For $q > 1/4$ this kurtosis exceeds 3. A standard normalized q-Gaussian mode would have fourth moment $2 + q$, so it cannot represent this complement.

6 Verification and the next calculation

The independent finite CAR driver uses integer fermion actions and Gaussian Wick sums, with no import of the old ladder driver. It evaluates all 81 fourth words and all nine second words for $(N, p) = (4, 1), (6, 1), (6, 2), (8, 2)$. In total it checks 972 pairing entries from 2,000,499 Gaussian pair-label assignments, plus all 116 sector-basis columns of the channel acting on one disjoint p-hop per fixture. It records six exact scalar ratio rows at $\lambda_N = 1/2$, $p = 2, 4, 8, 16, 32, 64$. Those asymptotic decimal rows are diagnostics, not interval certificates.

The first driver enumerates the $HHHH$ crossed contribution; it does not independently predict that contribution. A separate read-only scalar check compares it with the literature expression (12). A second independent symbolic audit uses the limiting two-edge rule to verify 62 identities, including the full 7×7 Gram matrix, (24), (29), and four exact finite residuals read from the CAR output. This checks algebraic consequences, not a replacement for the analytic limiting proof.

The scientific records are

- `results/susskind-slater-joint-fourth/exact-checks.json`;
- `results/susskind-slater-joint-fourth/gram-checks.json`;
- `results/susskind-slater-joint-fourth/covariance-checks.json`.

The stage evidence manifest binds source files, parameters, environment versions, parent evidence, build logs, PDF and page renders. Successful scientific outputs are preserved rather than rerun for document compilation.

The next target is the all-degree same-J joint law. At degree six, three simultaneously active labels can test whether the two-edge weights really factorize. A proof must retain general-hop overlaps, CAR signs, and backward selected pairings; pure-ladder suppression alone does not prove their suppression in the presence of H. Once those rules are established, an all-layer positive representation can supply the actual Hamiltonian evolution. The first-layer calculation here does not yet identify a de Sitter mass, position operator, stationary observer state, or finite-G gravitational four-point function. Those are subsequent physics questions, not substitutes for the joint-law calculation.

References

[1] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3 (2026).

[2] V. K. B. Kota, *SU(N) Wigner-Racah algebra for the matrix of second moments of embedded Gaussian unitary ensemble of random matrices*, arXiv:nucl-th/0410069 (2004).

[3] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev-Ye-Kitaev model in the double scaling limit*, arXiv:2006.13983v2 (2020).